

Slow-Roll Inflation in Cosmology

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1 Introduction

To explore the universe, people have constructed numerous models to fit the observation in astronomy. In 1981, Alan Guth[1] proposed a new concept “inflation” to resolve the horizon, flatness, and monopole problems. However, Guth’s inflation model introduced new issues. The most serious one could be the graceful-exit problem. Slow-roll inflation [2] is the most popular scenario which can get rid of this problem. We use natural units by setting $\hbar=c=8\pi G=1$

2 Friedmann equations

The line element ds in FLRW metric is

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu = -dt^2 + a^2(t)\left(\frac{dr^2}{1-kr^2} + r^2 d\theta^2 + r^2 \sin^2\theta d\varphi^2\right) \quad (1)$$

where $(x^0, x^1, x^2, x^3) = (t, r, \theta, \varphi)$

$a(t)$ is the scale factor and the parameter k corresponds to the shape of space. We take $k = 0$ for simplicity. First, we calculate the Christoffel symbol $\Gamma_{\mu\nu}^\alpha$:

$$\Gamma_{\mu\nu}^\alpha \equiv \frac{1}{2}g^{\alpha\beta}(\partial_\mu g_{\nu\beta} + \partial_\nu g_{\mu\beta} - \partial_\beta g_{\mu\nu}) \quad (2)$$

Using those Γ , we can derive the Riemann tensor $R_{\sigma\mu\nu}^\rho$:

$$R_{\sigma\mu\nu}^\rho \equiv \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda \quad (3)$$

We find that only the elements of $R_{\sigma\mu\nu}^\rho$ with $\rho = \mu$ and $\sigma = \nu$ are nonzero for FLRW metric. Then, the scalar curvature R can be obtained, which is

$$R = Tr_g(R_{\mu\nu}) = g^{\mu\nu} R_{\mu\nu} = \sum_{\mu=\nu=0}^3 g^{\mu\nu} R_{\mu\nu} = \frac{6}{a^2}(\dot{a}^2 + a\ddot{a}) \quad (4)$$

Now, we are ready to solve the Einstein field equations

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu} \quad (5)$$

where $G_{\mu\nu}$ is the Einstein tensor, $T_{\mu\nu}$ is the energy-momentum tensor of matter. According to observation, the universe is almost in thermal equilibrium, so we consider the matter as a perfect fluid, whose $T_{\mu\nu}$ has the form.

$$T_{\mu\nu} = (\epsilon + \frac{p}{c^2})u_\mu u_\nu + pg_{\mu\nu} \quad (6)$$

where ϵ is the mass-energy density, $u_\mu = \frac{dx_\mu}{dt}$ is the 4-velocity, and p is pressure.

The trace of $T_{\mu\nu}$ is

$$T = g^{\mu\nu}T_{\mu\nu} = (\epsilon + \frac{p}{c^2})g^{\mu\nu}u_\mu u_\nu + pg^{\mu\nu}g_{\mu\nu} = -\epsilon + 3p \quad (7)$$

Inserting the above T into the trace of Einstein equations (8), we get

$$(\frac{\dot{a}}{a})^2 + \frac{\ddot{a}}{a} = \frac{4\pi G}{3}(\epsilon - 3p) \quad (8)$$

Then, the 00-component of the Einstein equations () reads

$$H^2 = \frac{8\pi G}{3}\epsilon = \frac{1}{3}\epsilon \quad (9)$$

where $H = \dot{a}/a$ is the Hubble parameter.

Eq (10) is called the 2nd Friedmann equation. Substitute H^2 into $\frac{1}{3}\epsilon$ in eq (9), we find

$$\ddot{a} = -\frac{4\pi}{3}G(\epsilon + 3p)a \quad (10)$$

which is called the 1st Friedmann equation.

3 Slow-roll inflation

Given the action S :

$$S = \int \mathcal{L} d^4x = \int d^4x \sqrt{-g} \left[\frac{1}{2}R + \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi) \right] \quad (11)$$

which $\mathcal{L}$ is the Lagrangian density, $\sqrt{-g} = \sqrt{-\det g_{\mu\nu}}$. R is the scalar curvature. There are two independent dynamical variables in the action S . One is the scalar factor a , the other is the scalar field ϕ . Variation of S with respect to a and ϕ gives the Euler-Lagrange equations

$$H^2 = \frac{1}{3}(\frac{1}{2}\dot{\phi}^2 + V(\phi)) \quad (12)$$

$$\ddot{\phi} + 3H\dot{\phi} + \partial_\phi V(\phi) = 0 \quad (13)$$

Compare with eqs.(9),(10) and (12), we can see the energy density ϵ and pressure p

$$\epsilon = \frac{1}{2}\dot{\phi}^2 + V(\phi), \quad p = \frac{1}{2}\dot{\phi}^2 - V(\phi) \quad (14)$$

in this model.

“Slow-roll” means the universe acts like a particle slowly rolling down on a plateau of the potential $V(\phi)$ during the inflation. So, we assume that $V(\phi)$ is in the shape shown in figure 1: On the plateau, the value of $V(\phi)$ is much larger than the value of $\dot{\phi}$ in eq (12) while $\ddot{\phi}$ is negligible in eq (13). Then, under the slow-roll approximation, eq (12) and eq(13) become

$$H^2 \approx \frac{1}{3}V(\phi)$$

$$3H\dot{\phi} + \partial_{\phi}V(\phi) \approx 0$$
(15)

From the first equation above $H = \sqrt{\frac{V}{3}} = \frac{\dot{a}}{a}$ *constant*, we have

$$a = a_0 e^{Ht}$$
(16)

Which implies inflation

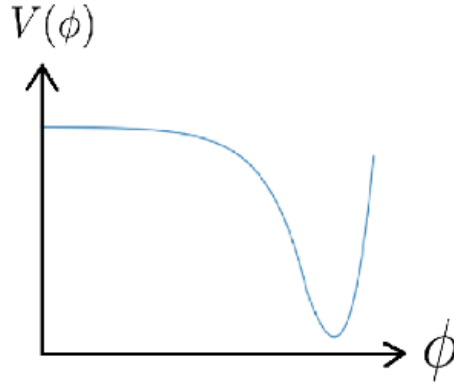


Figure 1

Reference

- [1] Alan H Guth, Eternal inflation and its implications, J. Phys. A: Math. Theor. 40 (2007) 6811–6826
- [2] A. Albrecht, P. J. Steinhardt, Cosmology for grand unified theories with radiatively induced symmetry breaking, Phys. Rev. Lett. 48, (1982) 1220